B. TECH-ME-III-I Sem

23A03504c	MECHANICAL BEHAVIOUR OF MATERIALS (Professional Elective-I)	L	T	P	C
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Course objectives: The objectives of the course are to	
1	Explain the structure of material over the effects of mechanical properties.
2	Familiarize the defects inside the structure and their effects on the mechanical properties.
3	Train the methods for characterization of the mechanical behavior of materials.
4	Impart knowledge about strengthening mechanisms of materials.
5	Teach mechanisms of failures of materials (fracture, fatigue and creep) and their relationship with the different types of stress.

COURSE OUTCOMES On successful completion of this course the student will be able		
CO1	Dictate the elastic behaviour of engineering materials, recall Hooke's law and apply the dislocation theory, forces on and between dislocations.	**L1, L2,L3**
CO2	Apply dispersion strengthening and fibre strengthening mechanisms, differentiate strain aging and dynamic strain aging and create grain size strengthening and solid solution strengthening	**L3, L4,L6**
CO3	List various modes of fracture and clarify the basic mechanism of ductile and brittle fracture, Identify importance of Griffith's theory. Calculate factors effecting on DBTT.	**L1,L2,L3, L6**
CO4	Explain fatigue behaviour and testing. Discuss the factors affecting fatigue. Apply fracture mechanics in design.	**L2, L3,L6**
CO5	Identify and describe various structural changes during creep. Evaluate and predict the metallurgical factors affecting creep and creep different testing.	**L2, L4,L5, L6**

UNIT – I

Elastic and plastic behavior: Elastic behavior of materials – Hooke's law, plastic behavior: dislocation theory – Burger's vectors and dislocation loops, dislocations in FCC, HCP and BCC lattice, stress fields and energies of dislocations, forces on and between dislocations, slip and twinning.

UNIT – II

Strengthening mechanisms: Cold Working, Grain Size Strengthening, Solid Solution Strengthening, Martensitic Strengthening, Precipitation Strengthening, Dispersion Strengthening, Fibre Strengthening, Examples. Yield Point Phenomenon, Strain aging and Dynamic strain aging.

UNIT – III

Fracture and fracture mechanics: Types of Fracture, Basic Mechanism of Ductile and Brittle Fracture, Griffith's Theory of Brittle Fracture, Ductile to Brittle Transition Temperature (DBTT), Factors Affecting DBTT, Determination of DBTT. Fracture Mechanics-Introduction, Modes of Fracture, Stress Intensity Factor, Strain Energy Release Rate, Fracture Toughness and Determination of KIC.

UNIT - IV

Fatigue behaviour and testing: Stress Cycles, S-N Curves, Effect of Mean Stress, Factors Affecting Fatigue, Structural Changes Accompanying Fatigue, Cumulative Damage, HCF / LCF, Thermo-mechanical Fatigue, Application of Fracture Mechanics to Fatigue Crack Propagation-Paris law- Fatigue Testing Machines.

UNIT - V

Creep behavior and testing: Creep Curve, Stages in Creep Curve and Explanation, Structural Changes during Creep, Creep Mechanisms, Metallurgical Factors Affecting Creep, High Temperature Alloys, Stress Rupture Testing, Creep Testing Machines.

Text books:

1. Dieter, G.E., "Mechanical Metallurgy", McGraw-Hill, SI Edition, 1995.
2. Davis. H. E., Troxell G.E., Hauck.G. E. W., "The Testing Of Engineering Materials", McGraw-Hill, 1982.

References:

1. Wulff, The Structure and Properties of Materials, Vol. III "Mechanical Behavior of Materials", John Wiley and Sons, 1983.
2. Honey Combe R. W. K., "Plastic Deformation of Materials", Edward Arnold Publishers, 1984.
3. Suryanarayana, A. V. K., "Testing of Metallic Materials", Prentice Hall India, 1979.

Online Learning Resources:

https://nptel.ac.in/courses/113104105

https://nptel.ac.in/courses/113104104

https://youtube.com/playlist?list=PLyqSpQzTE6M9QPU_tubmtQ97e7zRpaMlD&si=H5qNNyv3nYL8jztY

https://youtube.com/playlist?list=PLxQw8LdroTlPNimLKW-MWldJQHVLBESGs&si=ULCr6KGQwMPXhNC2

https://youtube.com/playlist?list=PL-g1KbXtGBBvF3G4lQuY0zSGBFHh4-5kF&si=47R1eQ_zAWcO-9A_